Suppose that I have an opaque jar with some number of ping pong balls in it. You know that the jar either has 10 ping pong balls in it or 1000 ping pong balls in it. In either case the balls are numbered sequentially. You think that the two possibilities are equally likely.

Suppose that we pick a ball at random. It is #3. Does this favor one hypothesis over the other?

It seems to favor the 10 ping pong ball hypothesis, since that hypothesis makes it more likely that #3 will be picked (rather than some number higher than 10).

Suppose someone offered you 40:1 odds that it is the 10-ball jar — so you win \$40 if it is the 1000 ball jar, and lose \$1 if it is the 10 ball jar. Should you take the bet?

But how much? How likely should I think that it is that the jar has 10 ping pong balls in it, vs. 1000?

The answer is not obvious. To answer this we need some rule which will tell us how to calculate probabilities in this kind of case.

What information do we have? We are comparing two different hypotheses: the 10-ball theory and the 1000-ball theory.

Before we drew the ball, we thought that each theory had a 0.5 probability of being correct. This is called the theory's **prior probability.**

We also know something else. We know how likely each theory predicts it to be that we draw the #3 ball.

We know that the 10-ball theory says that there is a 0.1 probability that the #3 ball will be drawn. The 1000-ball theory says that there is 0.001 probability that this ball will be drawn.

This is a claim about what is called **conditional probability**. This is the probability that something will happen if something else is true. What we seem to know here is the probability of the evidence (e) conditional on the hypothesis (h). We write that like this:

$$Pr(e \mid h)$$

So we know the prior probability of the hypotheses, and the probability of the evidence given the hypothesis. What we **want** to know is this:

$$Pr(h \mid e)$$

This is the probability of the hypothesis given the evidence. It tells us how likely we should take the hypothesis to be, given the evidence we have observed.

How can we do this?

One way to answer these questions employs a widely accepted rule of reasoning called "'Bayes' theorem," named after Thomas Bayes, an 18th century English mathematician and Presbyterian minister.

$$P(a|b) = \frac{P(a\&b)}{P(b)}$$

Here's an example to illustrate. Suppose that we have three candidates for political office. Mr. A has a 20% chance of winning, Mr. B has a 30% chance of winning, and Ms. C has a 50% chance of winning.

What's the probability that Mr. A wins given that a man wins?

fine-
tuning of the
universe

Bayes' theorem

the argument

objections

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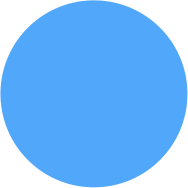

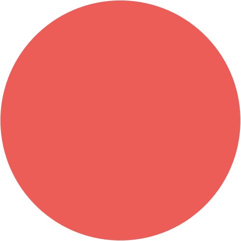

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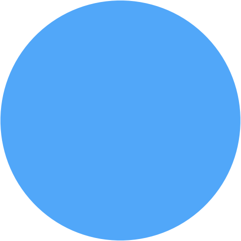

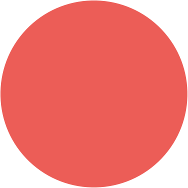

follows.

Enough about probability (for now). Let’s turn now to some results from contemporary physics which will be important to the argument which

A first question: what does the theory provided by physics include?

“The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.” (Hawthorne & Isaacs, ‘‘Fine-tuning fine-tuning’’)

These "dimensionless constants" will be our focus.

One such constant is the cosmological constant, which measures the energy density of empty space.

could have different values consistent with the laws of physics.

The cosmological constant is just a number which (as far as we know) likely it is, given the laws of the nature, that the fundamental constants

One thing that the standard model of physics gives us is a measure of how (like the cosmological constant) would fall in a certain range.

We can make certain plausible assumptions about what it would take for hard to see how life could have evolved. If there were no planets, it is hard to see how life could have evolved.

life to have evolved. For example, if there were nothing but hydrogen, it is of physics tells us about how likely it is that, for example, the cosmological

Given assumptions such as these, we can look at what the standard model constant has a value which would permit the evolution of life.

“Physicists have determined the (approximate) values of the fundamental constants by measurement. (There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

This is the claim that that the cosmological constant having a life-supporting value, given the laws of nature, is very low.

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. "Small" here doesn't mean "1 in 10,000" or "1 in 1,000,000". It means the kind of fraction that one would resort to exponents to describe, as in "1 in 10 to the 120". The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

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fine-
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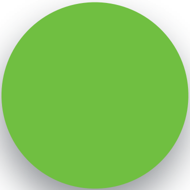

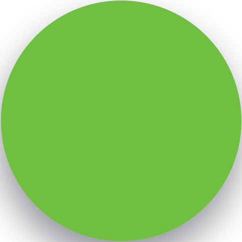

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The claim is that, according to current physics, the probability of the cosmological constant falling in a life-supporting range, given the laws of

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probability theory into an argument for the existence of God.

Let's now ask how to turn these facts about current physics and

Much as we considered two different hypotheses about the number of balls of the universe.

in the urn, so we can consider two different hypotheses about the origins created by an intelligent designer. Let the **non-design** hypotheses be the

Let the **design hypothesis** be the hypothesis that the universe was hypothesis that it was not.

Let our **evidence** (e) be the fact that the cosmological constant has a value in the life-permitting range.

Then what do we know about the relevant probabilities?

$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$

$$Pr(e \mid \text{design}) = 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h) * P(e|h)}{P(e)}$$

design hypothesis given our evidence is a matter of just plugging in the

With this information in hand, figuring out the probability of the nonnumbers.

$$Pr(\text{non-design}) = 0.5$$

$$Pr(\text{design}) = 0.5$$

$$Pr(e) = 0.25$$

$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

Powerball."

It is difficult to think about numbers as large as the denominator of this fraction. But consider the odds of winning Powerball one trillion times in a row. Call that a “super to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now that a “super duper Powerball.”

Now consider the odds of winning a super Powerball one trillion times in a row. Call odds of the universe being life-permitting by chance.

The odds of this happening are about $1 / 10^{44}$ — so much, much higher than the

Now consider the odds of winning a super duper Powerball one trillion times in a row.

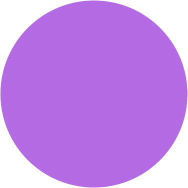
This means that if you simply take the physics at face value, and begin by assigning a the non-design hypothesis being true are vastly lower than the chances of winning a super duper powerball a trillion times in a row.

probability of 0.5 to the non-design hypothesis, you should think that the chances of minus the very small number we were just discussing:

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1

$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This is about as close to certainty as it is possible to get.



. (1,2,3,4)

1. P

2. P

6. Life exists.

5. P

7. $P(\text{design}) \approx 1$. (5,6)

8. If the universe was created by an

3. P(design)=P(non-design)=0.5.

4. Bayes' theorem.

C. P(God exists) $\approx$ 1. (7,8)

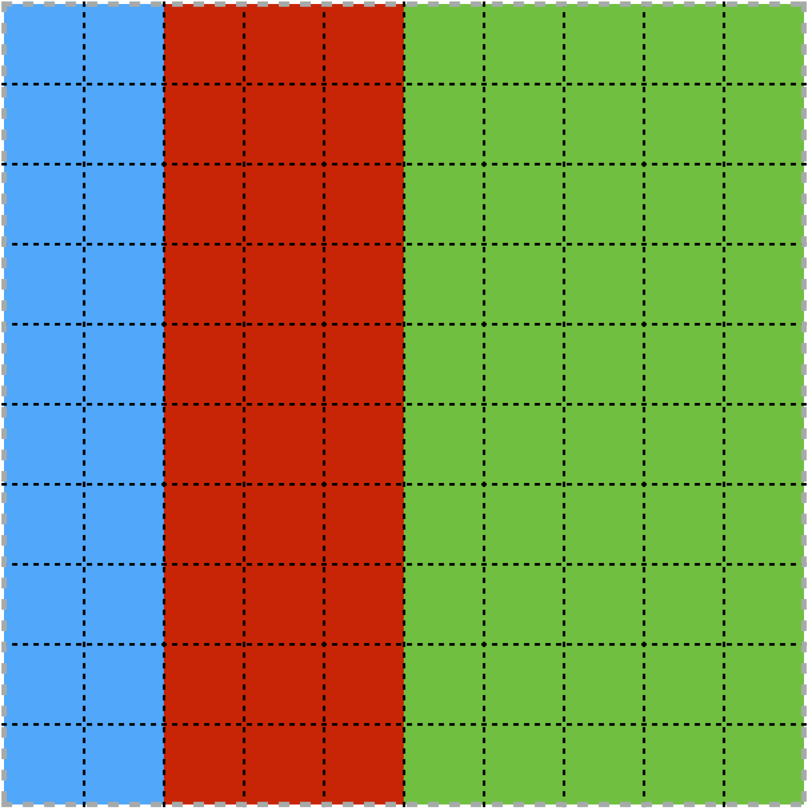
intelligent designer, then God exists.

$$(\texttt{life}|\texttt{design}) = 0.5$$

$$(\texttt{life}|\texttt{non-design}) = \frac{1}{10^{120}}$$

$$(\texttt{design}|\texttt{life}) \approx 1$$

Bayes'
theorem



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C

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